

① Find the area of the smaller region bounded by the curve $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and the line $\frac{x}{5} + \frac{y}{4} = 1$, using integration.

② Solve the following LPP graphically

Maximize : $Z = 3x + 3.5y$

Subject to conditions :

$$x + 2y \geq 240$$

$$3x + 1.5y \geq 270$$

$$1.5x + 2y \leq 310$$

$$x \geq 0, y \geq 0$$

③ Find $\int_{-1}^1 |x^4 - x| dx$

④ Find $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$

⑤ If A is a 2×3 matrix such that AB and AB' both are defined, then the order of the matrix B is

a) 2×2

b) 2×1

c) 3×2

d) 3×3

⑥ $\int_0^{\pi/4} \log(1 + \tan x) dx$

⑦ $\int \frac{dx}{\sqrt{\sin^3 x \cdot \cos(x-a)}}$

⑧ $\int e^{\cot^{-1} x} \left(\frac{1-x+x^2}{1+x^2} \right) dx$

⑨ $\int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx$